22MAT230 MATHEMATICS FOR COMPUTING 4

Batch : AIE -C

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LLM Guided Regularization of Pseudoinverse for Ill-Posed Signal Reconstruction

Table of Contents

LLM Guided Regularization of Pseudoinverse for Ill-Posed Signal Reconstruction 1
Introduction 2
Problem Statement 2
What Are ill Posed Inverse Problems? 3
Well Posed vs ill Posed Problems 3
- Full Rank Matrix 3
- ill Posed Matrix 4
  - Singular Value Spectrum 5
  - Stability Check 6
  - Condition Number via Singular 8
- Methodology 8

Signal Generation 9
Forward Models 11
- Gaussian Blur Operator 12
- Downsampling Operator 13
- Rank Deficient Operator 13
- Visuzalization Of Forward Operators 14
  - Noise Models 15
- Reconstruction Methods 16
  - PINV 17
  - TSVD 18
  - Tikhonov 20

NIST....21
FNIST....24
Evaluation Metrics....26
Model Comparison....26
LLM Parameter selection....27

Introduction

Many real-world systems do not allow direct observation of the true signal. Instead we observe corrupted measurements.

This leads to an inverse problem of the form:

$$y_\delta = \mathrm{Ax} + \eta$$

where

A : forward operator (system matrix)

x : true signal

y_δ : observed measurement

η : noise

Recovering x from noisy measurements y is difficult because many inverse problems are ill-posed and highly sensitive to noise.

Small perturbations in measurements can produce extremely large errors in the reconstructed signal.

To address this instability we study regularization methods such as:

1. Pseudoinverse (baseline)
2. Truncated SVD (TSVD)
3. Tikhonov Regularization
4. Non Stationary Iterated Tikhonov (NSIT)
5. Fast NSIT (FNIST)

In this project we further introduce an LLM guided parameter selection strategy to automatically recommend optimal regularization parameters.

Problem Statement

In this project, we study the reconstruction of an unknown signal from noisy measurements. The inverse problem is modeled as

$$\mathrm{Ax} = y_\delta$$

where

- A is the forward operator (system matrix)
- x is the unknown signal to be reconstructed

- y_δ represents the measured data.

In practice, the measured data contains noise and can be written as

$y_\delta = y + \eta$

where y is the exact noise free data and η represents measurement noise.

The noise level is assumed to satisfy $\| \eta \| \leq \delta$..

where δ denotes an upper bound on the magnitude of the noise present in the measurements.

The objective of the inverse problem is to estimate the unknown signal x from the noisy observations y_δ However, when the operator A is ill conditioned, direct inversion becomes unstable, making regularization methods necessary for stable reconstruction.

What Are ill Posed Inverse Problems?

A problem $\mathrm{Ax} = b$ is *WELL POSED* if:

1) Solution exists.

2) Solution is unique.

3) Solution depends continuously on the input data.

If any of these conditions are violated, the problem is called **ill-posed.**

Well Posed vs ill Posed Problems

We construct two matrices:

- a **full rank random matrix** representing a well-conditioned system
- a **rank deficient matrix** representing an ill-posed system.

The stability of each system is then analyzed using **Singular Value Decomposition (SVD)** and the **condition number**.

Full Rank Matrix

A **full rank Matrix** has all singular values as non zero.

For a well conditioned matrix, the smallest singular value is not close to zero.

This leads to

- moderate condition number
- stable inversion.

```
clc, clearvars
n = 100;
A1 = randn(n);              % Random Full rank matrix
```

```
[U1, S1, V1] = svd(A1);
singular_vals_A1 = diag(S1);

% Condition number
cond_A1 = cond(A1);
disp('Condtion number of Well Posed Matrix')
```

```
Condtion number of Well Posed Matrix
```

```
disp(cond_A1)
```

```
5.3544e+03
```

ill Posed Matrix

To simulate an ill-posed system, a **rank deficient operator** is constructed.

1. Random orthogonal matrices U and V are generated using SVD.
2. A singular value vector is defined as

$$s = [\sigma 1, \sigma 2, \ldots, \sigma r, 0, \ldots, 0]$$

where only the first r singular values are non-zero.

The matrix is constructed as

$A = U \Sigma V^T$

where Σ contains the predefined singular values.

Interpretation

Because several singular values are exactly zero, the matrix loses rank and becomes **rank deficient**.

This leads to:

- a very large condition number
- unstable inversion
- strong sensitivity to noise.

```
function A = rankdeficient_operator(n, r)
    [U,~,V] = svd(randn(n), 'econ');   % random orthogonal matrices
    s = [linspace(1,0.1,r) zeros(1,n-r)];   % r non-zero singular values
    A = U * diag(s) * V';   % construct matrix
end
A2 = rank_deficient_operator(n, 12);
[U2, S2, V2] = svd(A2);
singular_vals_A2 = diag(S2);

% Condition number
cond_A2 = cond(A2);
```

```
disp('Condition number of Ill posed Matrix:')
```

```
Condition number of Ill posed Matrix:
```

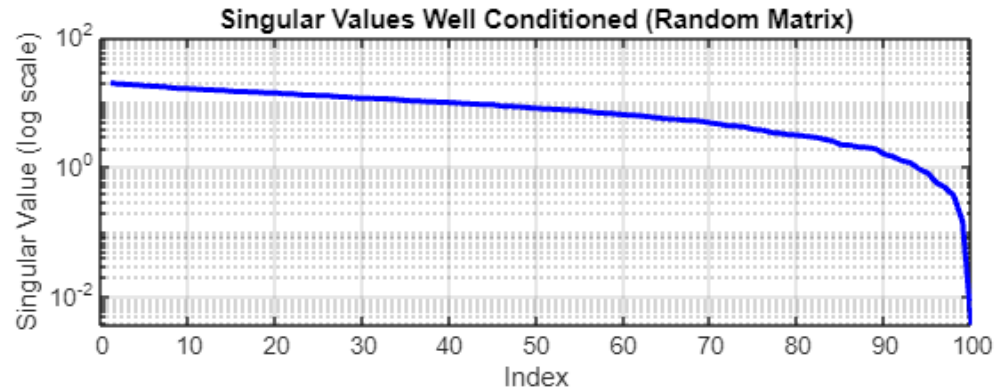
```
disp(cond_A2)
```

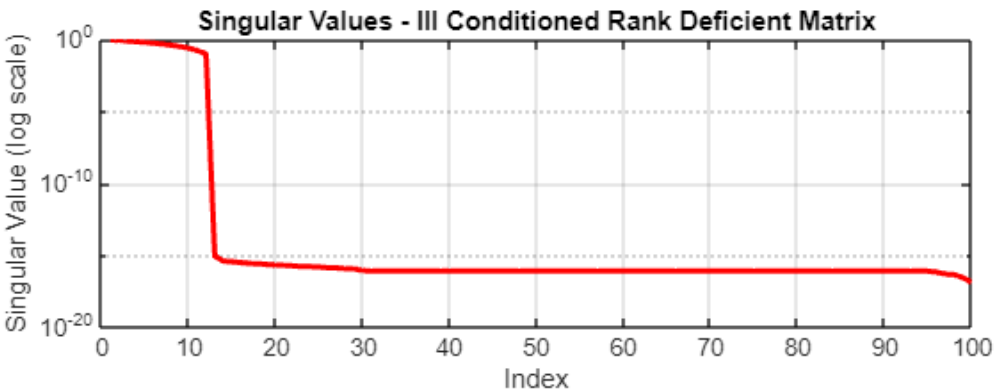
```
   3.1797e+18
```

Singular Value Spectrum

The singular values of Well-Conditioned and ill-posed Matrices are plotted on a **logarithmic scale** using a semilog plot.

```
figure;
subplot(2,1,1)
semilogy(singular_vals_A1, 'b', 'LineWidth', 2);
grid on;
title('Singular Values Well Conditioned (Random Matrix)');
xlabel('Index');
ylabel('Singular Value (log scale)');

subplot(2,1,2)
semilogy(singular_vals_A2, 'r', 'LineWidth', 2);
grid on;
title('Singular Values - Ill Conditioned Rank Deficient Matrix');
xlabel('Index');
ylabel('Singular Value (log scale)');
```





- The rapid decay of singular values causes instability in inverse problems. When attempting to compute the inverse solution, small singular values amplify noise present in the data.

For the random full-rank matrix:

- Singular values remain relatively well distributed.
- None of the singular values approach zero.
- The spectrum does not decay rapidly.

This indicates that the matrix is **well conditioned**, and inversion is numerically stable.

For the rank-deficient matrix:

- Singular values decay rapidly.
- Several singular values are exactly zero.
- The spectrum drops sharply.

This behavior indicates **ill conditioning** and loss of rank.

Stability Check

Two vectors are considered:

- $\mathrm{b1}$: the original data vector
- $\mathrm{b1}_{\mathrm{noisy}}$: a slightly perturbed version of the data

The noisy vector is generated by adding a very small random perturbation:

$\mathrm{b1}_{\mathrm{noisy}} = \mathrm{b1} + \varepsilon$

where ϵ is a small noise term.

The linear system is then solved using both vectors for the two matrices:

$x = A^{-1}\mathrm{b1}$

The difference between the resulting solutions is measured using the Euclidean norm:

$\| x_{\mathrm{original}} - x_{\mathrm{noisy}} \|$

Well Conditioned Matrix

For the full-rank matrix A_1:

- Small perturbations in the data produce **small changes in the solution**.
- The difference between the two solutions remains very small.
- This confirms that the system is **stable**.

Ill-Conditioned Matrix

For the rank-deficient matrix A_2:

- Even a tiny perturbation in the data causes **large variations in the solution**.
- The difference between the two solutions becomes significantly larger.
- This indicates that the system is **highly unstable**.

```
b1 = randn(n,1);
b1_noisy = b1 + 1e-6*randn(n,1);

x1_original = A1 \ b1;
x1_noisy    = A1 \ b1_noisy;
difference_well = norm(x1_original - x1_noisy);

x2_original = A2 \ b1;
```

```
Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND =  3.154765e-20.
```

```
x2_noisy    = A2 \ b1_noisy;
```

```
Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND =  3.154765e-20.
```

```
difference_ill = norm(x2_original - x2_noisy);

disp('Difference for Well-Conditioned Matrix:')
```

```
Difference for Well-Conditioned Matrix:
```

```
disp(difference_well)
```

```
   5.6937e-05
```

```

disp('Difference for Ill-Conditioned Matrix:')
```

```
Difference for Ill-Conditioned Matrix:
```

```
disp(difference_ill)
```

```
   4.1918e+12
```

Condition Number via Singular

For a matrix A, the condition number is defined as

$$\kappa\left(A\right) = \frac{\sigma_{\max}}{\sigma_{\min}}$$

where

- $\sigma_{\max}$ is the largest singular value of A
- $\sigma_{\min}$ is the smallest singular value of A

```
cond_manual_A1 = max(singular_vals_A1) / min(singular_vals_A1);
cond_manual_A2 = max(singular_vals_A2) / min(singular_vals_A2);

disp('Manual Condition Number (Random):')
```

```
Manual Condition Number (Random):
```

```
disp(cond_manual_A1)
```

```
   5.3544e+03
```

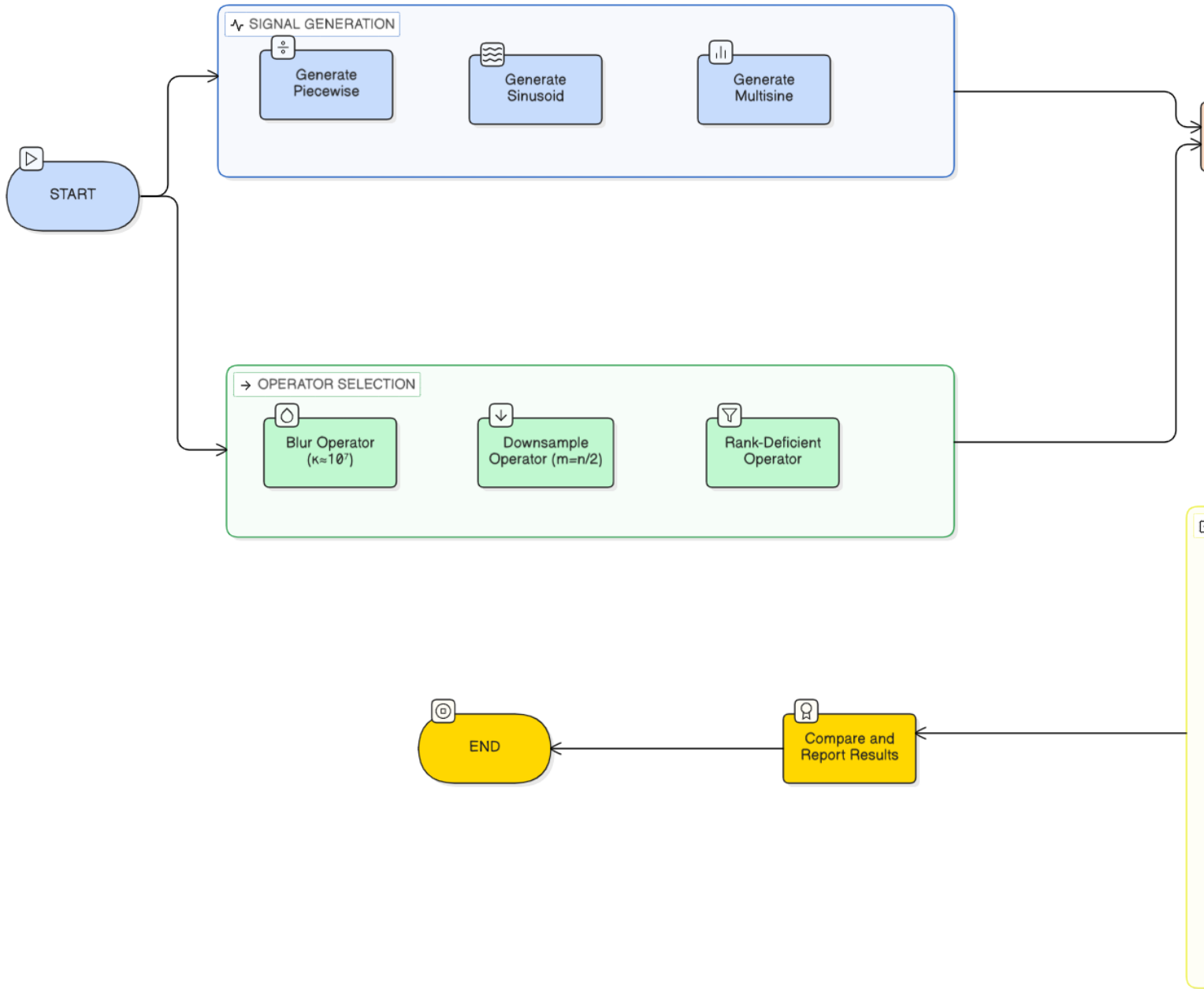
```

disp('Manual Condition Number (Hilbert):')
```

```
Manual Condition Number (Hilbert):
```

```
disp(cond_manual_A2)
```

```
   7.4082e+16
```

Methodology



Signal Generation

Sinusoidal Signal

The sinusoidal signal is defined as

$$y(t) = \sin(2\pi t) + 0.5\sin(6\pi t)$$

This represents a smooth periodic signal composed of two frequency components.

```
function y = sinusoid(t)
    y = sin(2*pi*t) + 0.5*sin(6*pi*t);
end
```

Multisine Signal

The multisine signal is constructed as the sum of multiple sinusoidal components

$$y(t) = \sum_{k=1}^{K} a_k \sin(2\pi f_k t)$$

where f_k represents the frequencies and a_k represents the amplitudes of the individual sine waves. This signal introduces more complex oscillatory behavior.

```
function y = multisine(t, freqs, amps)
    if nargin < 2
        freqs = [2 5 9];
    end
    if nargin < 3
        amps = [1.0 0.6 0.3];
    end

    y = zeros(size(t));
    for k = 1:length(freqs)
        y = y + amps(k) * sin(2*pi*freqs(k)*t);
    end
end
```

Piecewise Signal

The piecewise signal is defined as a signal with constant values over different intervals

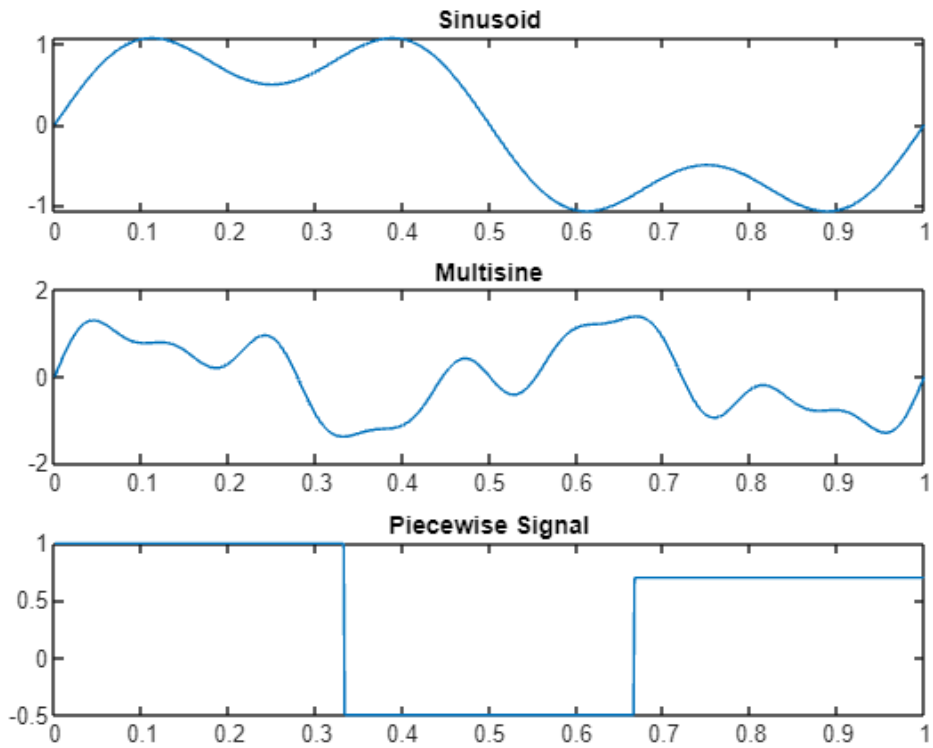
$$y(t) = \begin{cases} 1.0, & 0 \le t < \frac{1}{3} \\ -0.5, & \frac{1}{3} \le t < \frac{2}{3} \\ 0.7, & \frac{2}{3} \le t \le 1 \end{cases}$$

This signal contains abrupt transitions and discontinuities, making it useful for testing the ability of reconstruction methods to preserve sharp changes.

```
function y = piecewise_signal(t)
    y = zeros(size(t));
    N = length(t);
    thirds = floor(N/3);
```

```
    y(1:thirds) = 1.0;
    y(thirds+1:2*thirds) = -0.5;
    y(2*thirds+1:end) = 0.7;
end
```

```
t = linspace(0, 1, 1000);

y1 = sinusoid(t);
y2 = multisine(t);
y3 = piecewise_signal(t);

figure;
subplot(3,1,1); plot(t, y1); title('Sinusoid');
subplot(3,1,2); plot(t, y2); title('Multisine');
subplot(3,1,3); plot(t, y3); title('Piecewise Signal');
```



Forward Models

To simulate the measurement process in inverse problems, different **forward operators** are applied to the original signal. These operators model how the true signal is transformed before it is observed.

The general forward model is given by

$$y = \mathrm{A}\mathrm{x}$$

where A represents the forward operator, x is the original signal, and y is the observed measurement.

Gaussian Blur Operator

The Gaussian blur operator models a **smoothing process**, where the signal is convolved with a Gaussian kernel. The kernel determines how neighboring signal values contribute to the blurred measurement.

$$k(i) = \frac{1}{Z}\exp\left(-\frac{i^2}{2\sigma^2}\right)$$

where

- $k(i)$ - value of the Gaussian kernel at position iii
- σ - standard deviation controlling the amount of blur
- Z - normalization constant ensuring the kernel sums to 1.

The blurred measurement is obtained by applying the blur operator

$$y = A_{\text{blur}}x$$

where

- A_{blur} - matrix representation of the Gaussian blur operator
- x - original signal
- y - blurred measurement.

This operator suppresses high-frequency components of the signal.

```
function A = blur_operator(n, sigma, kernel_radius)
% BLUR_OPERATOR  Circulant Gaussian blur matrix

    if nargin < 3
        kernel_radius = 10;
    end

    ksize = 2 * kernel_radius + 1;
    k = gaussian_kernel(ksize, sigma);

    A = zeros(n, n);

    for i = 1:n
        for j = 1:n
            shift = mod(i - j, n);
            if shift < ksize
                A(i, j) = k(shift + 1);
            end
        end
    end
```

```
end
function k = gaussian_kernel(sz, sigma) % GAUSSIAN_KERNEL 1D Gaussian kernel
    idx = -(floor(sz/2)) : floor(sz/2);
    k = exp(-0.5 * (idx / sigma).^2);
    k = k / sum(k);
end
```

Downsampling Operator

The downsampling operator simulates the situation where measurements are taken at a **lower resolution**. Instead of observing every element of the signal, only selected samples are recorded.

Each measurement is obtained as

$$y_i = x_{(i-1)f+1}$$

where

- y_i : ith measurement value
- $x(j)$: value of the original signal at index j
- f downsampling factor determining the sampling interval.

In matrix form the model becomes

$$y = A_{\text{down}} x$$

where

- A_{down} : downsampling operator matrix
- x : original signal
- y : reduced measurement vector.

This operator causes **loss of information**, making the inverse problem more difficult.

```
function A = downsample_operator(n, factor)
% DOWNSAMPLE_OPERATOR  Downsampling matrix

    if nargin < 2
        factor = 2;
    end

    m = floor(n / factor);
    A = zeros(m, n);

    for i = 1:m
        A(i, (i-1)*factor + 1) = 1.0;
    end
end
```

Rank Deficient Operator

A rank-deficient operator is constructed using the **Singular Value Decomposition (SVD)** to simulate an ill-posed system.

Equation

$$A = U\Sigma V^T$$

where

- U : orthogonal matrix containing left singular vectors
- V : orthogonal matrix containing right singular vectors
- Σ : diagonal matrix containing the singular values.

The singular value matrix is defined as

Equation

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 & \cdots \\ 0 & \sigma_2 & 0 & \cdots \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \sigma_r \end{bmatrix}$$

where

- $\sigma 1, \sigma 2, \ldots, \sigma r$ are non-zero singular values
- remaining entries are zero.

Since only the first r singular values are non-zero, the matrix has reduced rank

$$\mathrm{rank}(A) = r < n$$

where

- r : rank of the matrix
- n : dimension of the matrix.

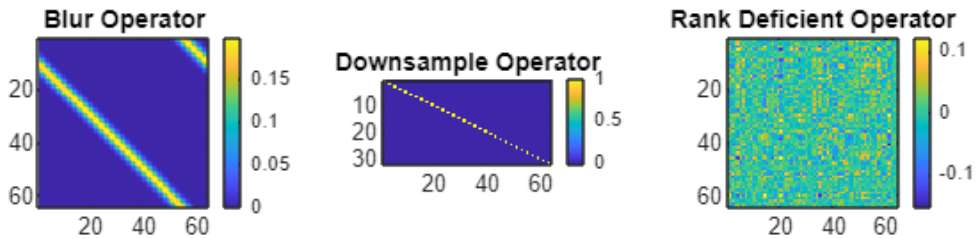
This rank deficiency makes the system **ill-posed**, because certain components of the signal cannot be recovered.

```
function A = rank_deficient_operator(n, r)
    [U,~,V] = svd(randn(n), 'econ');   % random orthogonal matrices
    s = [linspace(1,0.1,r) zeros(1,n-r)];   % r non-zero singular values
    A = U * diag(s) * V';   % construct matrix
end
```

Visuzalization Of Forward Operators

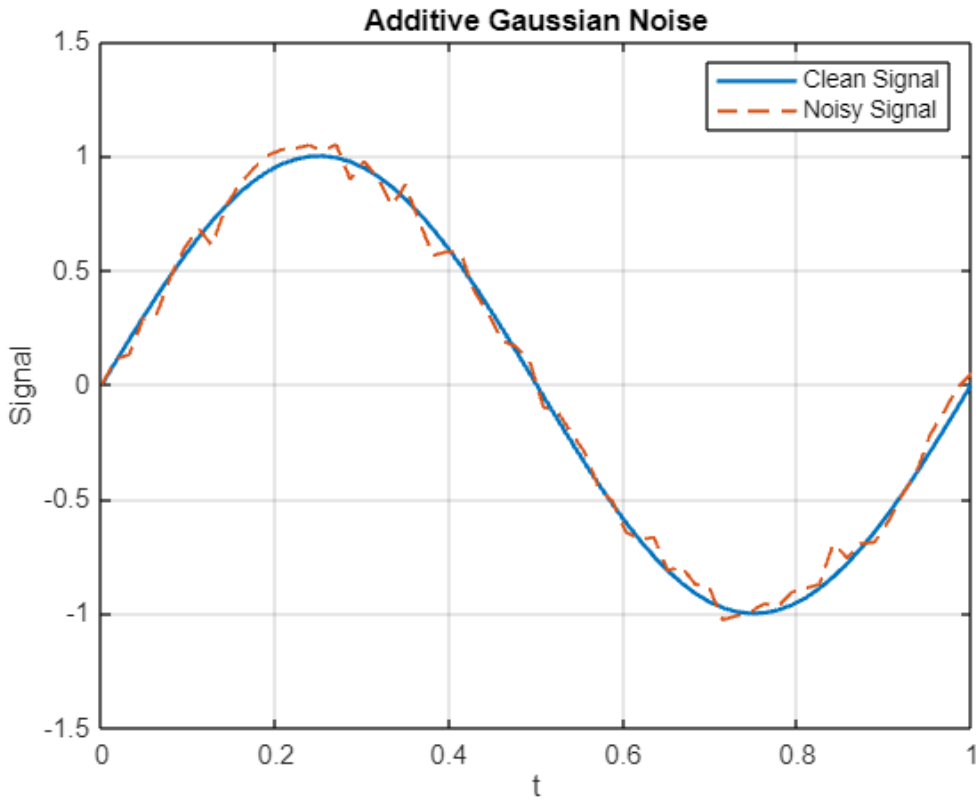
```
n = 100;

A_blur = blur_operator(n, 2.0);
```

```
A_down = downsample_operator(n, 2);
A_rank = rankdeficient_operator(n, 12);

figure('Position',[100 100 1000 300])
subplot(1,3,1); imagesc(A_blur); axis image;
title('Blur Operator'); colorbar;

subplot(1,3,2); imagesc(A_down); axis image;
title('Downsample Operator'); colorbar;

subplot(1,3,3); imagesc(A_rank); axis image;
title('Rank Deficient Operator'); colorbar;
```



Noise Models

```
n = 100;
t = linspace(0,1,n)';
x = sin(2*pi*t);

sigma = 0.05;   % noise level

[y_noisy, noise] = add_gaussian_noise(x, sigma);

figure
plot(t,x,'LineWidth',1.5)
hold on
plot(t,y_noisy,'--','LineWidth',1.2)

legend('Clean Signal','Noisy Signal')
title('Additive Gaussian Noise')
xlabel('t')
ylabel('Signal')
grid on
```




```
function [y_noisy, noise] = add_gaussian_noise(y, sigma, rng)

    if nargin < 3 || isempty(rng)
        noise = sigma * randn(size(y));
    else
        noise = sigma * randn(rng, size(y));
    end

    y_noisy = y + noise;

end
```

Reconstruction Methods

After constructing the forward blur operator A, several reconstruction techniques are applied to recover the original signal x from the noisy observation y. where:

- A – Forward operator (blur matrix)
- x – True signal
- y – Observed noisy signal
- η – Additive noise

The goal of reconstruction is to estimate x from y.

```
n = 100;              % signal length
noise_level = 0.08;   % noise standard deviation
blur_sigma = 2.5;     % blur kernel width

% Generate true signal
t = linspace(0,1,n);
x_true = sin(2*pi*t)' + 0.5*sin(6*pi*t)';

% Create forward operator (blur matrix)
A = blur_operator(n, blur_sigma, 10);

% Compute condition number
cond_num = cond(A);

fprintf('Forward operator matrix A: shape = %d x %d\n', size(A,1), size(A,2));
```

```
Forward operator matrix A: shape = 100 x 100
```

```
fprintf('Condition number (using diagnostics): κ(A) = %.2e\n', cond_num);
```

```
Condition number (using diagnostics): κ(A) = 1.94e+06
```

```


% Forward model
y_clean = A * x_true;

% Add Gaussian noise
[y_noisy, noise_vec] = add_gaussian_noise(y_clean, noise_level);

%From this we will use A and y_noisy for reconstruction
```

PINV

The pseudoinverse method computes the least-squares solution using the Moore Penrose pseudoinverse.

$x = A^{\dagger} y$ where $A^{\dagger}$ is the pseudoinverse of A.

Characteristics

- Provides the minimum norm solution.
- Works well when the matrix is well-conditioned.
- Highly sensitive to noise in ill-posed problems.

```
function x = pseudoinverse_reconstruct(A, y_noisy)
    % Reconstruct signal using pseudoinverse
    % Inputs:
    % A - Forward operator matrix (n x m)
    % y_noisy - Noisy measurement vector (n x 1)

    % Output:
    % x - Reconstructed signal (m x 1)
```

```
    x = pinv(A) * y_noisy;
end

% Reconstruct
x_pinv = pseudoinverse_reconstruct(A, y_noisy);

% Compute reconstruction errors
mse_val = mse_error(x_true, x_pinv);
rel_error = relative_error(x_true, x_pinv);

fprintf('Relative Error: %.4f\n', rel_error);
```

```
Relative Error: 27107.9725
```

```
fprintf('MSE: %.6f\n', mse_val);
```

```
MSE: 454683593.547804
```

TSVD

TSVD stabilizes the inversion by truncating small singular values of the forward operator.

$A = U\Sigma V^T$

Only the first k largest singular values are used: $x_k = V_k \Sigma_k^{-1} U_k^T y$

Characteristics

- Reduces amplification of noise.
- Regularization controlled by truncation parameter k.
- Smaller k increases stability but may lose signal detail.

Parameter Selection

A parameter sweep is performed over possible values of k to determine the optimal reconstruction based on minimum MSE.

```
function x = tsvd_reconstruct(A, y, k)
    % TSVD_RECONSTRUCT  Truncated SVD reconstruction
    % Inputs:
    % A - Forward operator matrix
    % y - noisy observation
    % k - truncation parameter
    % Output:
    % x - reconstructed signal

    [U,S,V] = svd(A,'econ');

    % Truncate SVD components
    Uk = U(:,1:k);
    Sk = S(1:k,1:k);
```

```
    Vk = V(:,1:k);

    % TSVD solution
    x = Vk * (Sk \ (Uk' * y));
end
```

```
% Parameter sweep for TSVD
step = max(1, floor(n/50));
ks = 1:step:n-1;

num_k = length(ks);

mse_vals = zeros(num_k,1);
rel_vals = zeros(num_k,1);

x_store = cell(num_k,1);

for i = 1:num_k

    k = ks(i);

    % TSVD reconstruction
    x_tsvd = tsvd_reconstruct(A, y_noisy, k);

    % Metrics
    mse_vals(i) = mse_error(x_true, x_tsvd);
    rel_vals(i) = relative_error(x_true, x_tsvd);

    x_store{i} = x_tsvd;

end

% Find optimal k
[best_mse_tsvd, best_idx] = min(mse_vals);
best_k = ks(best_idx);
best_rel_tsvd = rel_vals(best_idx);
x_best_tsvd = x_store{best_idx};

fprintf('\nTSVD Parameter Sweep Results:\n');
```

```
TSVD Parameter Sweep Results:
```

```
fprintf('k range: %d to %d\n', ks(1), ks(end));
```

```
k range: 1 to 99
```

```
fprintf('Optimal k: %d\n', best_k);
```

```
Optimal k: 7
```

```
fprintf('Optimal MSE: %.6f\n', best_mse_tsvd);
```

```
Optimal MSE: 0.000948
```

```
fprintf('Optimal Relative Error: %.6f\n', best_rel_tsvd);
```

```
Optimal Relative Error: 0.039142
```

Tikhonov

Tikhonov regularization stabilizes the inverse problem by adding a penalty term. The objective function of Tikhonov regularization is $\min_x (\|Ax - y\|^2 + \lambda\|x\|^2)$

$$J(x) = (Ax - y)^T(Ax - y) + \lambda x^T x$$

$$J(x) = x^T A^T Ax - 2y^T Ax + y^T y + \lambda x^T x$$

$$\frac{\partial J(x)}{\partial x} = 2A^T Ax - 2A^T y + 2\lambda x$$

$$2A^T Ax - 2A^T y + 2\lambda x = 0$$

$$A^T Ax + \lambda x = A^T y$$

$$(A^T A + \lambda I)x = A^T y$$

The closed form solution is: $x = (A^T A + \lambda I)^{-1} A^T y$

```
function x_tik = tikhonov_reconstruct(A, y, lambda)
% Tikhonov reconstruction
% Solves: min ||Ax - y||^2 + lambda ||x||^2

% Inputs:
% A - Forward operator matrix
% y - Observed noisy data
% lambda - Regularization parameter

% Output:
% x_tik - Reconstructed signal

[m, n] = size(A);

% Identity matrix
I = eye(n);

% Tikhonov solution
x_tik = (A' * A + lambda * I) \ (A' * y);
end
```

```
lambdas = logspace(-6,2,50);

mse_vals = zeros(length(lambdas),1);
```

```
rel_err_vals = zeros(length(lambdas),1);

x_results = cell(length(lambdas),1);

for i = 1:length(lambdas)

    lam = lambdas(i);

    % Reconstruction
    x_tik = tikhonov_reconstruct(A, y_noisy, lam);

    % Metrics
    mse_vals(i) = mse_error(x_true, x_tik);
    rel_err_vals(i) = relative_error(x_true, x_tik);

    x_results{i} = x_tik;

end

% Find optimal lambda
[best_mse_tikh, best_idx] = min(mse_vals);

best_lambda = lambdas(best_idx);
best_rel_tikh = rel_err_vals(best_idx);
x_best_tikh = x_results{best_idx};

fprintf('Tikhonov Parameter Sweep Results:\n');
```

```
Tikhonov Parameter Sweep Results:
```

```
fprintf('Lambda range: %.2e to %.2e\n', lambdas(1), lambdas(end));
```

```
Lambda range: 1.00e-06 to 1.00e+02
```

```
fprintf('Optimal lambda: %.6e\n', best_lambda);
```

```
Optimal lambda: 7.906043e-02
```

```
fprintf('Optimal MSE: %.6f\n', best_mse_tikh);
```

```
Optimal MSE: 0.008548
```

```
fprintf('Optimal Relative Error: %.6f\n', best_rel_tikh);
```

```
Optimal Relative Error: 0.117539
```

NIST

NSIT is an iterative regularization technique where the regularization parameter decreases geometrically during iterations.

To determine when to stop the iterations, the algorithm uses the **Morozov Discrepancy Principle**.

The measurement model is assumed to be $y = Ax_{\text{true}} + \epsilon$ where A is the forward operator, x_{true} is the true signal, and ϵ represents measurement noise. The noise level is assumed to be bounded by $\|\epsilon\| \leq \delta$ where δ represents the known noise magnitude.

During the reconstruction process, the algorithm computes the **residual** $r_k = Ax_k - y$ which measures the difference between the predicted data Ax_k and the observed data y. The residual norm $\|r_k\| = \|Ax_k - y\|$ indicates how well the reconstructed solution explains the measurements.

According to the Morozov discrepancy principle, the reconstruction should not attempt to fit the noise. Therefore, the iterations are stopped when the residual becomes approximately equal to the noise level, $\|Ax_k - y\| \approx \delta$

In practice, a small tolerance factor τ>1, is used to allow a slight margin, $\|Ax_k - y\| \leq \tau\delta$.

Thus, the algorithm stops at the smallest iteration k for which this condition is satisfied. This ensures that the reconstruction fits the data up to the expected noise level without overfitting the noise.

```
function [x_rec, residuals, rel_errors, stop_iter] = nsit_morozov(A, y, x_true,
delta, q, tau, max_iter)

% NSIT reconstruction using Morozov discrepancy principle
% A        : forward operator
% y        : noisy measurements
% x_true   : true signal (only for error evaluation)
% delta    : noise level
% q        : geometric decay factor (0<q<1)
% tau      : Morozov safety factor
% max_iter : maximum iterations

[m,n] = size(A);

x = zeros(n,1);

At = A';
AtA = At*A;

% Proper scaling of alpha
alpha0 = norm(A)^2;

residuals = zeros(max_iter,1);
rel_errors = zeros(max_iter,1);

for k = 1:max_iter

    % Geometric alpha schedule
    alpha = alpha0 * q^(k-1);

    % Compute residual
    r = y - A*x;
```

```
    % NSIT correction step
    correction = (AtA + alpha*eye(n)) \ (At*r);

    % Update solution
    x = x + correction;

    % Residual norm
    residuals(k) = norm(y - A*x);

    % Relative error (for evaluation)
    rel_errors(k) = relative_error(x_true, x);

    % Morozov stopping rule
    if residuals(k) <= tau * delta
        break
    end

end

x_rec = x;
stop_iter = k;

residuals = residuals(1:k);
rel_errors = rel_errors(1:k);

end
```

```
max_iter = 200;
q = 0.9;
tau = 1.01;

delta = norm(noise_vec);

[x_nsit, residuals, errors, stop_iter] = nsit_morozov(A, y_noisy, x_true, delta, q,
tau, max_iter);

nsit_best_mse = mse_error(x_true,x_nsit);
nsit_best_rel = errors(end);

fprintf('NSIT stopped at iteration %d\n', stop_iter);
```

```
NSIT stopped at iteration 5
```

```
fprintf('Mean Squared Error = %.6f\n', nsit_best_mse);
```

```
Mean Squared Error = 0.003386
```

```
fprintf('Relative Error = %.6f\n', nsit_best_rel);
```

```
Relative Error = 0.073978
```

FNIST

FNIST is an iterative regularization method where the regularization parameter decreases geometrically during iterations. This helps stabilize the inverse problem while gradually improving the reconstruction accuracy.

$x_0 = 0$ The reconstruction starts with an initial estimate of the solution. $\alpha_k = \alpha_0 q^{k-1}$ where

- α0 is the initial regularization parameter
- q∈(0,1) is the decay factor
- k is the iteration index.

$r_k = y - Ax_k$ The residual measures the difference between observed data and the predicted data.

$F_1 = (A^T A + \alpha_k I)^{-1} A^T r_k$ This term is similar to a **Tikhonov update** and moves the solution toward reducing the residual.

$F_2 = \alpha_k (A^T A + \alpha_k I)^{-1} (A^T A + \alpha_k^2 I)^{-1} A^T r_k$ This second correction improves stability and compensates for approximation errors.

Update rule: $x_{k+1} = x_k + 0.7F_1 + 0.3F_2$ The two correction terms are combined with weights to obtain a stable iterative update.

$x_{\text{rec}} = x_{k_{\text{final}}}$ After the specified number of iterations, the final solution is taken as the reconstructed signal.

$\alpha_0 \in [10^{-2}, 10^2], \quad q \in [0.85, 0.99], \quad k \in [5, 30]$ After the specified number of iterations, the final solution is taken as the reconstructed signal.

This gradually reduces the regularization strength.

```
function x_rec = fnist_reconstruction(A, y, alpha0, q, max_iter)
    [~,n] = size(A);
    x = zeros(n,1);
    At = A';
    AtA = At*A;

    for k = 1:max_iter
        alpha = alpha0 * q^(k-1);
        r = y - A*x;
        % First correction
        F1 = (AtA + alpha*eye(n)) \ (At*r);
        % Second correction (correct form)
        temp = (AtA + alpha^2*eye(n)) \ (At*r);
        F2 = alpha * ((AtA + alpha*eye(n)) \ temp);
        % Stable update
        x = x + 0.7*F1 + 0.3*F2;
    end
    x_rec = x;
end
```

```
alpha0 = 1;
q = 0.9;
max_iter = 10;

x_fnist = fnist_reconstruction(A, y_noisy, alpha0, q, max_iter);

% metrics
mse_val = mse_error(x_true, x_fnist);
rel_error = relative_error(x_true, x_fnist);

fprintf('\nFNIST Reconstruction Results:\n');
```

```
FNIST Reconstruction Results:
```

```
fprintf('Relative Error: %.6f\n', rel_error);
```

```
Relative Error: 0.124670
```

```
fprintf('MSE: %.6f\n', mse_val);
```

```
MSE: 0.009617
```

```


alpha_vals = logspace(-2,2,25);   % alpha0 values
q_vals = 0.85:0.01:0.99;           % decay factors
iter_vals = 5:30;                 % iterations

best_mse = inf;
best_rel = inf;
best_alpha = 0;
best_q = 0;
best_iter = 0;
best_x = zeros(size(x_true));

for a = 1:length(alpha_vals)
    for b = 1:length(q_vals)
        for c = 1:length(iter_vals)

            alpha0 = alpha_vals(a);
            q = q_vals(b);
            max_iter = iter_vals(c);

            x_fnist = fnist_reconstruction(A, y_noisy, alpha0, q, max_iter);

            mse_val = mse_error(x_true, x_fnist);
            rel_err = relative_error(x_true, x_fnist);

            if mse_val < best_mse
                best_mse = mse_val;
                best_rel = rel_err;
                best_alpha = alpha0;
                best_q = q;
```

```
                best_iter = max_iter;
                best_x = x_fnist;
            end
        end
    end
end

best_mse_fnist = best_mse;
best_rel_fnist = best_rel;
x_fnist_best = best_x;

fprintf("\nOptimal FNIST Parameters:\n");
fprintf("alpha0 = %.4f\n", best_alpha);
fprintf("q = %.3f\n", best_q);
fprintf("iterations = %d\n", best_iter);
fprintf("Best MSE = %.6f\n", best_mse_fnist);
fprintf("Best Relative Error = %.6f\n", best_rel_fnist);
```

Evaluation Metrics

$$\text{Relative Error} = \frac{\|x_{\text{true}} - x_{\text{rec}}\|}{\|x_{\text{true}}\|}$$

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(x_{\text{true},i} - x_{\text{rec},i})^2$$

```
function err = relative_error(x_true, x_rec)
err = norm(x_true - x_rec) / norm(x_true);
end

function mse = mse_error(x_true, x_rec)
mse = mean((x_true - x_rec).^2);
end
```

Model Comparison

This section compares all the methods by its Evaluation metrics.

```
methods = {'Pseudoinverse','TSVD','Tikhonov','NSIT','FNIST'};

rel_errors = [
    relative_error(x_true, x_pinv);
    best_rel_tsvd;
    best_rel_tikh;
    nsit_best_rel;
    best_rel_fnist
];

mses = [
    mse_error(x_true, x_pinv);
```

```
    best_mse_tsvd;
    best_mse_tikh;
    nsit_best_mse;
    best_mse_fnist
];

fprintf('\n=====================================\n');
```

```
=====================================
```

```
fprintf('Reconstruction Method Comparison\n');
```

```
Reconstruction Method Comparison
```

```
fprintf('=====================================\n');
```

```
=====================================
```

```
for i = 1:length(methods)
    fprintf('%-12s | Rel Error: %.6f | MSE: %.6f\n',...
        methods{i}, rel_errors(i), mses(i));
end
```

```
Pseudoinverse | Rel Error: 27107.972469 | MSE: 454683593.547804
TSVD          | Rel Error: 0.039142 | MSE: 0.000948
Tikhonov      | Rel Error: 0.117539 | MSE: 0.008548
NSIT          | Rel Error: 0.073978 | MSE: 0.003386
FNIST         | Rel Error: 0.065126 | MSE: 0.002624
```

LLM Parameter selection

This section builds prompt for feeding to the LLM, and this prompt contains all the necessary information like signal type, signal length, matrix size , rank , condition number and the errors obtained from all the methods. Specifying the format (json) of the output and from the response we get back the required parameters and reconstruct with the signal with that.

```
% BUILD PROMPT FOR LLM

prompt = "";

prompt = prompt + "You are an expert in numerical linear algebra and inverse
problems." + newline;
prompt = prompt + "Your task is to recommend optimal regularization parameters for
solving an ill-posed inverse problem." + newline;
prompt = prompt + "We are solving the reconstruction problem Ax = y where A is
ill-conditioned and the measurements contain noise." + newline + newline;

prompt = prompt + "=====================================" + newline;
prompt = prompt + "PROBLEM DESCRIPTION" + newline;
prompt = prompt + "=====================================" + newline;

prompt = prompt + "Signal model:" + newline;
```

```
prompt = prompt + "- Signal type: multisine signal (sum of multiple sinusoids)" +
newline;
prompt = prompt + "- Signal length: " + num2str(n) + newline;
prompt = prompt + "- Sampling interval: uniform in [0,1]" + newline + newline;

prompt = prompt + "Forward model:" + newline;
prompt = prompt + "- Operator type: Gaussian blur operator" + newline;
prompt = prompt + "- Blur sigma: " + num2str(blur_sigma) + newline;
prompt = prompt + "- Matrix size: " + num2str(size(A,1)) + " x " +
num2str(size(A,2)) + newline;
prompt = prompt + "- Matrix rank: " + num2str(rank(A)) + newline;
prompt = prompt + "- Condition number: " + num2str(cond_num) + newline + newline;

prompt = prompt + "Noise model:" + newline;
prompt = prompt + "- Additive Gaussian noise" + newline;
prompt = prompt + "- Noise standard deviation: " + num2str(noise_level) + newline;
prompt = prompt + "- Noise norm: " + num2str(norm(noise_vec)) + newline + newline;

prompt = prompt + "=====================================" + newline;
prompt = prompt + "CURRENT RECONSTRUCTION PERFORMANCE" + newline;
prompt = prompt + "=====================================" + newline;

prompt = prompt + "Relative reconstruction errors:" + newline;
prompt = prompt + "- Pseudoinverse: " + num2str(rel_errors(1)) + newline;
prompt = prompt + "- TSVD: " + num2str(rel_errors(2)) + newline;
prompt = prompt + "- Tikhonov: " + num2str(rel_errors(3)) + newline;
prompt = prompt + "- NSIT: " + num2str(rel_errors(4)) + newline;
prompt = prompt + "- FNIST: " + num2str(rel_errors(5)) + newline + newline;

prompt = prompt + "Mean squared errors:" + newline;
prompt = prompt + "- Pseudoinverse: " + num2str(mses(1)) + newline;
prompt = prompt + "- TSVD: " + num2str(mses(2)) + newline;
prompt = prompt + "- Tikhonov: " + num2str(mses(3)) + newline;
prompt = prompt + "- NSIT: " + num2str(mses(4)) + newline;
prompt = prompt + "- FNIST: " + num2str(mses(5)) + newline + newline;

prompt = prompt + "=====================================" + newline;
prompt = prompt + "OBJECTIVE" + newline;
prompt = prompt + "=====================================" + newline;

prompt = prompt + "Select improved parameters that minimize reconstruction error
and improve stability." + newline;
prompt = prompt + "Consider the ill-conditioning of A and the noise level when
choosing parameters." + newline;
prompt = prompt + "Avoid parameter choices that cause instability or noise
amplification." + newline + newline;

prompt = prompt + "=====================================" + newline;
prompt = prompt + "ALGORITHMS TO TUNE" + newline;
prompt = prompt + "=====================================" + newline;
```

```
prompt = prompt + "1) TSVD: choose truncation parameter k" + newline;
prompt = prompt + "2) Tikhonov: choose regularization parameter lambda" + newline;
prompt = prompt + "3) NSIT: choose q, iterations, delta, tau" + newline;
prompt = prompt + "4) FNIST: choose alpha0, q, iterations" + newline + newline;

prompt = prompt + "======================================" + newline;
prompt = prompt + "PARAMETER CONSTRAINTS" + newline;
prompt = prompt + "======================================" + newline;

prompt = prompt + "- k must be between 5 and 30" + newline;
prompt = prompt + "- lambda between 1e-6 and 1" + newline;
prompt = prompt + "- alpha0 between 0.001 and 0.5" + newline;
prompt = prompt + "- q between 0.5 and 0.95" + newline;
prompt = prompt + "- iterations between 10 and 60" + newline;
prompt = prompt + "- tau between 1.0 and 1.2" + newline;
prompt = prompt + "- delta should be close to the noise norm" + newline + newline;

prompt = prompt + "======================================" + newline;
prompt = prompt + "RESPONSE FORMAT" + newline;
prompt = prompt + "======================================" + newline;

prompt = prompt + "Return ONLY valid JSON with no explanation." + newline;
prompt = prompt + "Use exactly this structure:" + newline;

prompt = prompt + '{' + newline;
prompt = prompt + ' "TSVD": {' + newline;
prompt = prompt + '   "k": 12' + newline;
prompt = prompt + ' },' + newline;
prompt = prompt + ' "Tikhonov": {' + newline;
prompt = prompt + '   "lambda": 0.015' + newline;
prompt = prompt + ' },' + newline;
prompt = prompt + ' "NSIT": {' + newline;
prompt = prompt + '   "q": 0.92,' + newline;
prompt = prompt + '   "iterations": 15,' + newline;
prompt = prompt + '   "delta": 0.1,' + newline;
prompt = prompt + '   "tau": 1.02' + newline;
prompt = prompt + ' },' + newline;
prompt = prompt + ' "FNIST": {' + newline;
prompt = prompt + '   "alpha0": 0.15,' + newline;
prompt = prompt + '   "q": 0.9,' + newline;
prompt = prompt + '   "iterations": 18' + newline;
prompt = prompt + ' }' + newline;
prompt = prompt + '}';
```

```
api_key = "YOUR_API_KEY_HERE";
params = get_llm_parameters_gemini(prompt, api_key);
k_llm = params.TSVD.k;
```

```
lambda_llm = params.Tikhonov.lambda;

q_nsit = params.NSIT.q;
iter_nsit = params.NSIT.iterations;
delta_nsit = params.NSIT.delta;
tau_nsit = params.NSIT.tau;

alpha_fnist = params.FNIST.alpha0;
q_fnist = params.FNIST.q;
iter_fnist = params.FNIST.iterations;

disp("LLM Suggested Parameters:")
```

```
LLM Suggested Parameters:
```

```
disp(params)
```

```
        TSVD: [1×1 struct]
    Tikhonov: [1×1 struct]
        NSIT: [1×1 struct]
       FNIST: [1×1 struct]
```

```
% RECONSTRUCTION WITH LLM PARAMETERS

% Tikhonov
x_rec_tikh_llm = tikhonov_reconstruct(A, y_noisy, lambda_llm);

% TSVD
x_rec_tsvd_llm = tsvd_reconstruct(A, y_noisy, k_llm);

% NSIT
[x_rec_nsit_llm,~,~,~] =
nsit_morozov(A,y_noisy,x_true,delta_nsit,q_nsit,tau_nsit,iter_nsit);

% FNIST
x_rec_fnist_llm = fnist_reconstruction( ...
    A, ...
    y_noisy, ...
    alpha_fnist, ...
    q_fnist, ...
    iter_fnist );

% COMPUTE ERRORS
rel_tsvd_llm  = relative_error(x_true, x_rec_tsvd_llm);
rel_nsit_llm  = relative_error(x_true, x_rec_nsit_llm);
rel_fnist_llm = relative_error(x_true, x_rec_fnist_llm);
rel_tikh_llm = relative_error(x_true, x_rec_tikh_llm);

mse_tsvd_llm  = mse_error(x_true, x_rec_tsvd_llm);
mse_nsit_llm  = mse_error(x_true, x_rec_nsit_llm);
mse_fnist_llm = mse_error(x_true, x_rec_fnist_llm);
```

```
mse_tikh_llm = mse_error(x_true, x_rec_tikh_llm);
```

```
% VISUALIZATION

figure

subplot(3,2,1)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',y_noisy,'--')
title("Original vs Observed")
legend("True","Observed")
grid on

subplot(3,2,2)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',x_rec_tsvd_llm)
title("TSVD")
legend("True","TSVD","TSVD + LLM")
```

```
Warning: Ignoring extra legend entries.
```

```
grid on

subplot(3,2,3)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',x_rec_tikh_llm)
title("Tikhonov")
legend("True","Tikhonov","Tikhonov + LLM")
```

```
Warning: Ignoring extra legend entries.
```

```
grid on

subplot(3,2,4)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',x_rec_nsit_llm)
title("NSIT")
legend("True","NSIT","NSIT + LLM")
```

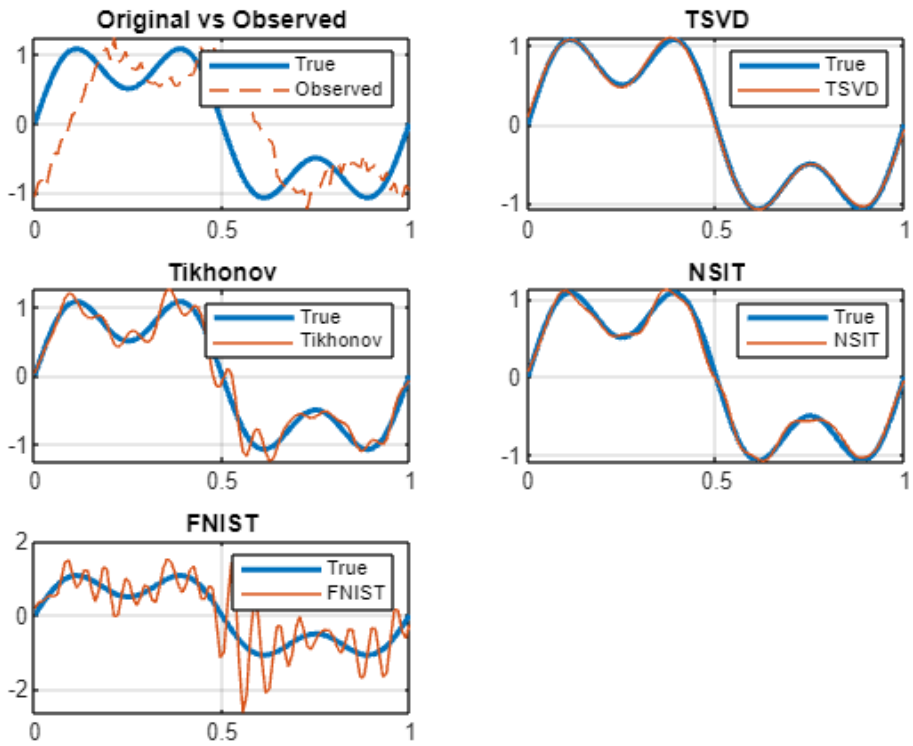
```
Warning: Ignoring extra legend entries.
```

```
grid on

subplot(3,2,5)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',x_rec_fnist_llm)
```

```
title("FNIST")
legend("True","FNIST","FNIST + LLM")
```

```
Warning: Ignoring extra legend entries.
```

```
grid on
```




```
% PRINT FINAL RESULTS

fprintf("\n----- Reconstruction Errors -----\n")
```

```
----- Reconstruction Errors -----
```

```
fprintf("TSVD                : %.6f\n",best_rel_tsvd)
```

```
TSVD              : 0.039142
```

```
fprintf("TSVD + LLM          : %.6f\n",rel_tsvd_llm)
```

```
TSVD + LLM        : 0.041002
```

```
fprintf("Tikhonov            : %.6f\n",best_rel_tikh)
```

```
Tikhonov          : 0.117539
```

```
fprintf("Tikhonov + LLM      : %.6f\n",rel_tikh_llm)
```

```
Tikhonov + LLM    : 0.181156
```

```
fprintf("NSIT                : %.6f\n",nsit_best_rel)
```

```
NSIT             : 0.073978
```

```
fprintf("NSIT + LLM          : %.6f\n",rel_nsit_llm)
```

```
NSIT + LLM       : 0.073509
```

```
fprintf("FNIST               : %.6f\n",best_rel_fnist)
```

```
FNIST            : 0.065126
```

```
fprintf("FNIST + LLM         : %.6f\n",rel_fnist_llm)
```

```
FNIST + LLM      : 0.700339
```

```
function params = get_llm_parameters_gemini(prompt, api_key)

url = "https://generativelanguage.googleapis.com/v1/models/gemini-2.5-
flash:generateContent?key=" + api_key;
data = struct;
data.contents = {struct("parts",{struct("text",prompt)})};

options = weboptions( ...
    'MediaType','application/json', ...
    'ContentType','json', ...
    'Timeout',60);

response = webwrite(url,data,options);

text = response.candidates(1).content.parts(1).text;
% Remove markdown if present
text = erase(text,"```json");
text = erase(text,"```");

% Extract JSON safely
start_idx = find(text=='{',1,'first');
end_idx   = find(text=='}',1,'last');

json_text = text(start_idx:end_idx);

params = jsondecode(json_text);

end
```